

# Homework #4

Math 104: Intro to Analysis

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Donald Aingworth IV

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## 0 Various Theorems

### 0.1 Limit Proof I

For  $p, q, \alpha, C \in \mathbb{R}$  where  $\alpha > 1$  and  $C > 0$ ,  $\lim_{q \rightarrow p} C |p - q|^{\alpha-1} = 0$ .

*Proof.* Let  $\varepsilon > 0$ . Let  $\delta = \alpha^{-1} \sqrt[\alpha]{\frac{\varepsilon}{C}}$ , which is greater than zero because  $\alpha - 1 > 0$  and  $C > 0$ . Suppose that  $0 < |p - q| < \delta$ . Substitute in our value of  $\delta$  to find  $0 < |p - q| < \alpha^{-1} \sqrt[\alpha]{\frac{\varepsilon}{C}}$ . Now, raise all sides to the positive power  $\alpha - 1$ , then multiply by  $C$ .

$$0 < |p - q| < \alpha^{-1} \sqrt[\alpha]{\frac{\varepsilon}{C}} \quad (1)$$

$$0 < |p - q|^{\alpha-1} < \frac{\varepsilon}{C} \quad (2)$$

$$0 < C |p - q|^{\alpha-1} < \varepsilon \quad (3)$$

Since  $0 < C |p - q|^{\alpha-1}$ ,  $C |p - q|^{\alpha-1} = \left| C |p - q|^{\alpha-1} \right|$ . Therefore,  $\left| C |p - q|^{\alpha-1} \right| < \varepsilon$ . Ergo,  $\lim_{q \rightarrow p} C |p - q|^{\alpha-1} = 0$ . OEΔ

### 0.2 Limit Proof II

$$\lim_{x \rightarrow \infty} \frac{1}{x} = 0 \quad (4)$$

*Proof.* Let  $\varepsilon > 0$ . Let  $\delta = \frac{1}{\varepsilon} > 0$ . Suppose that  $x > \delta$ . Multiply both sides by the reciprocals of  $x$  and  $\delta$  to find that  $\frac{1}{\delta} > \frac{1}{x}$ . Substitute in the value of  $\delta$  to find that  $\frac{1}{\frac{1}{\varepsilon}} = \varepsilon > \frac{1}{x}$ . Since  $\delta > 0$  and  $x > \delta$ ,  $x > 0$ , meaning  $\frac{1}{x} > 0$ , so  $\frac{1}{x} = \left| \frac{1}{x} \right|$ . So  $\varepsilon > \left| \frac{1}{x} \right|$ . Therefore,  $x > \delta$  implies  $\varepsilon > \left| \frac{1}{x} \right|$ . Ergo,  $\lim_{x \rightarrow \infty} \frac{1}{x} = 0$ . OEΔ

### 0.3 Alternate Derivative Definition

Let  $f : \mathbb{R} \rightarrow \mathbb{R}$  be differentiable.

$$\frac{df(x)}{dx} = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \quad (5)$$

*Proof.* Recall the definition of  $\frac{df(x)}{dx}$ .

$$\frac{df(x)}{dx} = \lim_{c \rightarrow x} \frac{f(c) - f(x)}{c - x} \quad (6)$$

This means that for all  $\varepsilon > 0$ , there exists  $\delta > 0$  such that  $|c - x| < \delta$  implies  $|f(c) - f(x)| < \varepsilon$ . Let  $h = c - x$ . Solving for  $c$ ,  $c = x + h$ . We can therefore conclude that  $|h| < \delta$  implies  $|f(x+h) - f(x)| < \varepsilon$ . This fits a definition of a new limit.

$$\frac{df(x)}{dx} = \lim_{c \rightarrow x} \frac{f(c) - f(x)}{c - x} = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \quad (7)$$

This completes the proof. OEΔ

## 0.4 Alternate Mean Value Theorem

Let  $f$  be a real, continuous function on  $[x, x+h]$  which is differentiable in  $(x, x+h)$ , then there is a point  $b \in (x, x+h)$  at which

$$f(x+h) - f(x) = hf'(b). \quad (8)$$

*Proof.* Recall the Mean Value Theorem. If  $f$  is a real continuous function on  $[x, y]$  which is differentiable in  $(x, y)$ , then there is a point  $b \in (x, y)$  at which

$$f(y) - f(x) = (y-x)f'(b) \quad (9)$$

Let  $h = y - x$ . By extension,  $y = x + h$ . Therefore, for  $b \in (x, x+h)$ ,

$$f(x+h) - f(x) = hf'(b) \quad (10)$$

OEΔ

Let  $f$  be a real, continuous function on  $[x-h, x]$  which is differentiable in  $(x-h, x)$ , then there is a point  $c \in (x-h, x)$  at which

$$f(x) - f(x-h) = hf'(c). \quad (11)$$

*Proof.* This is very similar to the last. Recall the Mean Value Theorem. If  $f$  is a real continuous function on  $[y, x]$  which is differentiable in  $(y, x)$ , then there is a point  $c \in (y, x)$  at which

$$f(x) - f(y) = (x-y)f'(c) \quad (12)$$

Let  $h = x - y$ . By extension,  $y = x - h$ . Therefore, for  $c \in (x-h, x)$ ,

$$f(x) - f(x-h) = hf'(c) \quad (13)$$

OEΔ

Note that these are respectively equivalent to:

$$f'(b) = \frac{f(x+h) - f(x)}{h} \quad (14)$$

$$f'(c) = \frac{f(x) - f(x-h)}{h} \quad (15)$$

## 0.5 Mean in the Middle

For  $a, b \in \mathbb{R}$ , if  $a < b$ , then  $a < \frac{a+b}{2} < b$ .

*Proof.* Let  $a, b \in \mathbb{R}$  where  $a < b$ . Add  $b$  to both sides and find  $a+b < 2b$ . Divide both sides by two to find  $\frac{a+b}{2} < b$ .

Take  $a < b$ . Add  $a$  to both sides and find  $2a < a+b$ . Divide both sides by two to find  $a < \frac{a+b}{2}$ .

Ergo, by the transitive property,  $a < \frac{a+b}{2} < b$ . OEΔ

# 1 Problem 1

Let  $(X, d_X)$  and  $(Y, d_Y)$  be metric spaces. A function  $f : (X, d_X) \rightarrow (Y, d_Y)$  is called *Lipschitz continuous* if there is a constant  $L > 0$  so that

$$d_Y(f(p), f(q)) \leq L d_X(p, q)$$

for all  $p, q \in X$ . For  $\alpha > 0$ , we say  $f$  is *Hölder continuous (with exponent  $\alpha$ )* if there is a constant  $C > 0$  so that

$$d_Y(f(p), f(q)) \leq C d_X(p, q)^\alpha$$

for all  $p, q \in X$ . In particular, Hölder continuous with exponent 1 means the same thing as Lipschitz.

- (a) Fix a point  $p \in X$ . Show that the map  $g : X \rightarrow \mathbb{R}$  defined by  $g(x) = d(x, p)$  is Lipschitz continuous, with constant  $L = 1$ .
- (b) Prove that any Hölder continuous function (with any exponent  $\alpha > 0$ ) is uniformly continuous.
- (c) Prove that if  $f$  is Lipschitz continuous and bounded, then  $f$  is Hölder continuous with exponent  $\alpha$ , for any  $0 < \alpha \leq 1$ .
- (d) Let  $[a, b]$  be a closed interval, and suppose  $f : [a, b] \rightarrow \mathbb{R}$  is  $C^1$  (meaning its first derivative exists and is continuous on  $[a, b]$ ). Show that  $f$  is Lipschitz continuous.
- (e) Suppose  $f : \mathbb{R} \rightarrow \mathbb{R}$  is Hölder continuous with exponent  $\alpha > 1$ . Show that  $f$  is constant.

## 1.1 Solution (a)

*Proof.* Since the codomain of  $g$  is  $\mathbb{R}$ , we can use the standard metric as  $d_Y$ . From the triangle inequality, we know that  $d_X(x, y) + d_X(p, x) \geq d_X(p, y)$  and  $d_X(x, y) + d_X(p, y) \geq d_X(p, x)$ . We can reorder both of these to find that  $d_X(x, y) \geq d_X(p, y) - d_X(p, x)$  and  $d_X(x, y) \geq d_X(p, x) - d_X(p, y)$ . Consolidating these, we can apply the absolute value to find that  $d_X(x, y) \geq |d_X(p, y) - d_X(p, x)|$ . This is also known as the inverse triangle inequality. Since the absolute value is the usual metric over  $\mathbb{R}$ , this means that  $d_X(x, y) \geq d_{\mathbb{R}}(d_X(p, y), d_X(p, x))$ . Since  $g(x) = d_X(p, x)$ ,  $d_X(x, y) \geq d_{\mathbb{R}}(g(y), g(x))$ . Therefore,  $g$  is Lipschitz continuous with constant  $L = 1$ . OEΔ

## 1.2 Solution (b)

*Proof.* Let  $f$  be Hölder continuous with exponent  $\alpha$  for  $\alpha > 0$ .

$$d_Y(f(p), f(q)) \leq C d_X(p, q)^\alpha \tag{16}$$

Let  $\varepsilon > 0$  and let  $\delta = \sqrt[\alpha]{\frac{\varepsilon}{C}}$ , which allows that  $\delta > 0$  since  $C > 0$ . Suppose that for some  $p, q \in X$ ,  $d_X(p, q) < \delta$ . We can substitute in our value of  $\delta$  and isolate  $\varepsilon$ .

$$d_X(p, q) < \delta = \sqrt[\alpha]{\frac{\varepsilon}{C}} \tag{17}$$

$$d_X(p, q)^\alpha < \frac{\varepsilon}{C} \tag{18}$$

$$C d_X(p, q)^\alpha < \varepsilon \tag{19}$$

Since  $d_Y(f(p), f(q)) \leq C d_X(p, q)^\alpha$ , this means  $d_Y(f(p), f(q)) < \varepsilon$ . Therefore,  $d_X(p, q) < \delta$  implies  $d_Y(f(p), f(q)) < \varepsilon$ . Ergo,  $f$  is uniformly continuous. OEΔ

### 1.3 Solution (c)

*Proof.* Let  $0 < \alpha \leq 1$ , and let  $f$  be Lipschitz continuous. Let  $p, q \in X$ .

$$d_Y(f(p), f(q)) \leq L d_X(p, q) \quad (20)$$

Since  $f$  is bounded, then there exists a number  $M \in \mathbb{R}$  and a point  $f(q) \in Y$  for some  $q \in X$  such that for all  $p \in X$ ,  $d_Y(f(p), f(q)) < M$ . Suppose we take three cases:  $d_X(p, q) = 1$ ,  $d_X(p, q) > 1$ , and  $d_X(p, q) < 1$ .

**Case 1:**  $d_X(p, q) = 1$ . Anything real to the power of 1 is just 1, so if we set  $C = M$ , then it will always be true that  $d_Y(f(p), f(q)) \leq C d_X(p, q)^\alpha$ .

**Case 2:**  $d_X(p, q) < 1$ . If  $d_X(p, q) < 1$ , then for  $0 < \alpha \leq 1$ ,  $d_X(p, q) \leq d_X(p, q)^\alpha$ . Therefore, if we set  $C = L$ , then  $L d_X(p, q) \leq L d_X(p, q)^\alpha$ , and therefore  $d_Y(f(p), f(q)) \leq C d_X(p, q)^\alpha$ .

**Case 3:**  $d_X(p, q) > 1$ . If  $d_X(p, q) > 1$ , then for all  $0 < \alpha \leq 1$ ,  $d_X(p, q)^\alpha > 1$ . Therefore, if we set  $C = M$ , then  $C d_X(p, q)^\alpha > M$ , and by extension it will always be true that  $d_Y(f(p), f(q)) < C d_X(p, q)^\alpha$ .

Therefore, depending on the value of our inserts, we can take  $C = \max\{M, L\}$  as a constant and  $f$  will be Hölder continuous. OEΔ

### 1.4 Solution (d)

*Proof.*  $[a, b]$  is a subset of the real numbers, so we will be using the usual metric. If  $f$  is  $C^1$ , then its first derivative exists and is continuous on  $[a, b]$ . Intervals are compact in the real numbers, so  $[a, b]$  is compact. Since continuous functions on compact sets are bounded,  $f'([a, b])$  is bounded. There exists  $M \in \mathbb{R}$  such that for all  $x \in [a, b]$ ,  $|f'(x)| < M$ . Without loss of generality, suppose  $p \leq q$ . By the mean value theorem, there exists  $c \in (p, q)$  so that  $f'(c) = \frac{f(q) - f(p)}{q - p}$ . Since  $c \in [a, b]$ ,  $|f'(x)| < M$ , so  $\left| \frac{f(q) - f(p)}{q - p} \right| < M$ . By the limit laws,  $\frac{|f(q) - f(p)|}{|q - p|} < M$ . Multiply both sides by  $|q - p|$  to find that  $|f(q) - f(p)| < M |q - p|$ . Since  $\mathbb{R}$  uses the usual metric, this means for any  $x, p \in X$ ,  $d_{\mathbb{R}}(f(p), f(q)) \leq L d_{[a, b]}(p, q)$  for any  $p, q \in [a, b]$ . Therefore,  $f$  is Lipschitz continuous. OEΔ

### 1.5 Solution (e)

*Proof.* Let  $f$  be Hölder continuous with exponent  $\alpha > 1$ . There exists a constant  $C$  such that for all  $p, q \in \mathbb{R}$ :

$$d_{\mathbb{R}}(f(p), f(q)) \leq C d_{\mathbb{R}}(p, q)^\alpha \quad (21)$$

Note that  $\mathbb{R}$  uses the usual metric.

$$|f(p) - f(q)| \leq C |p - q|^\alpha \quad (22)$$

Divide both sides by  $|p - q|$ .

$$\frac{|f(p) - f(q)|}{|p - q|} = \left| \frac{f(p) - f(q)}{p - q} \right| \leq C |p - q|^{\alpha-1} \quad (23)$$

By the definition of the absolute value,  $\left| \frac{f(p) - f(q)}{p - q} \right| \geq 0$ .

$$0 \leq \left| \frac{f(p) - f(q)}{p - q} \right| \leq C |p - q|^{\alpha-1} \quad (24)$$

Take the limit of all sides as  $q$  approaches  $p$ .

$$\lim_{q \rightarrow p} 0 \leq \lim_{q \rightarrow p} \left| \frac{f(p) - f(q)}{p - q} \right| \leq \lim_{q \rightarrow p} C |p - q|^{\alpha-1} \quad (25)$$

We proved in *0.1: Limit Proof I* that  $\lim_{q \rightarrow p} C |p - q|^{\alpha-1} = 0$ . Since 0 is constant and contains neither  $p$  nor  $q$ ,  $\lim_{q \rightarrow p} 0 = 0$ . Therefore, by the squeeze theorem,  $\lim_{q \rightarrow p} \left| \frac{f(p) - f(q)}{p - q} \right| = f'(p) = 0$  for all  $p \in \mathbb{R}$ .

Now, recall the mean value theorem: if  $f$  is a real, continuous function on  $[a, b]$  which is differentiable on  $(a, b)$ , there is a point  $x \in (a, b)$  where  $f(b) - f(a) = (b - a)f'(x)$ . Since our  $f'(x) = 0$  for all  $x \in \mathbb{R}$ , fix  $p$  and let  $q \in \mathbb{R}$ . The Mean Value Theorem tells us that for the interval  $[p, q]$ , there exists a point  $x \in \mathbb{R}$  such that  $f(q) - f(p) = (q - p)f'(x) = (q - p) * 0 = 0$ . Therefore, for any  $q \in \mathbb{R}$ ,  $f(p) = f(q)$ . Ergo,  $f$  is constant. OEΔ

## 2 Problem 2

On the interval  $(100, \infty)$ , define functions  $f(x) = x + \sin(x)$ ,  $g(x) = x$ ,  $p(x) = x + \sin(x) \cos(x)$ , and  $q(x) = p(x)e^{\sin(x)}$ .

(a) Using facts we proved about limits and derivatives, prove that

$$\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = 1, \quad \lim_{x \rightarrow \infty} \frac{f'(x)}{g'(x)} \text{ DNE}, \quad \lim_{x \rightarrow \infty} \frac{p(x)}{q(x)} \text{ DNE}, \quad \text{and} \quad \lim_{x \rightarrow \infty} \frac{p'(x)}{q'(x)} = 0.$$

(b) For both scenarios  $\frac{f(x)}{g(x)}$  and  $\frac{p(x)}{q(x)}$ , explain why these results do not contradict L'Hôpital's rule.

*Note: even though we haven't learned about sin and cos yet, you are free to use any familiar properties of them you wish.*

### 2.1 Solution (a)

We prove that  $\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = 1$ .

*Proof.* First write out the limit.

$$\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = \lim_{x \rightarrow \infty} \frac{x + \sin(x)}{x} = \lim_{x \rightarrow \infty} \left[ 1 + \frac{\sin(x)}{x} \right] \quad (26)$$

By limit laws, we can separated added up properties.

$$\lim_{x \rightarrow \infty} \left[ 1 + \frac{\sin(x)}{x} \right] = \lim_{x \rightarrow \infty} 1 + \lim_{x \rightarrow \infty} \frac{\sin(x)}{x} \quad (27)$$

Since 1 is constant,  $\lim_{x \rightarrow \infty} 1 = 1$ . My first assertion about sin: it is bounded to  $[-1, 1]$  and is cyclical, so  $-1 \leq \sin(x) \leq 1$ . Divide all sides by  $x$  to find  $-\frac{1}{x} \leq \frac{\sin(x)}{x} \leq \frac{1}{x}$ . Take the limit as  $x$  approaches  $\infty$  to get  $-\lim_{x \rightarrow \infty} \frac{1}{x} = \lim_{x \rightarrow \infty} -\frac{1}{x} \leq \lim_{x \rightarrow \infty} \frac{\sin(x)}{x} \leq \lim_{x \rightarrow \infty} \frac{1}{x}$ . We proved earlier that  $\lim_{x \rightarrow \infty} \frac{1}{x} = 0$ , and by extension,  $-\lim_{x \rightarrow \infty} \frac{1}{x} = 0$ . Therefore, by the sequence squeeze theorem,  $\lim_{x \rightarrow \infty} \frac{\sin(x)}{x} = 0$ . Put this back into  $\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)}$ .

$$\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = \lim_{x \rightarrow \infty} 1 + \lim_{x \rightarrow \infty} \frac{\sin(x)}{x} = 1 + 0 = 1 \quad (28)$$

OEΔ

We prove that  $\lim_{x \rightarrow \infty} \frac{f'(x)}{g'(x)}$  does not exist.

*Proof.* My second assertion is that if  $f(x) = \sin(x)$ ,  $f'(x) = \cos(x)$ . Since  $(a + b)' = a' + b'$ ,  $(x + \sin(x))' = (x)' + (\sin(x))' = 1 + \cos(x)$ .

$$\lim_{x \rightarrow \infty} \frac{f'(x)}{g'(x)} = \lim_{x \rightarrow \infty} [1 + \cos(x)] \quad (29)$$

Create two sequences in terms of  $n \in \mathbb{N}$  that diverge to  $\infty$  as  $n \rightarrow \infty$ .

$$a_n = 2n\pi; b_n = (2n + 1)\pi \quad (30)$$

Apply both  $a_n$  and  $b_n$  to  $1 + \cos(x)$ .

$$1 + \cos(a_n) = 1 + \cos(2n\pi) = 1 + 1 = 2 \quad (31)$$

$$1 + \cos(b_n) = 1 + \cos((2n+1)\pi) = 1 - 1 = 0 \quad (32)$$

This gives us two subsequences that converge to two different values as  $x \rightarrow \infty$ . Therefore, the limit does not exist. OEΔ

We prove that  $\lim_{x \rightarrow \infty} \frac{p(x)}{q(x)}$  does not exist.

*Proof.*

$$\lim_{x \rightarrow \infty} \frac{p(x)}{q(x)} = \lim_{x \rightarrow \infty} \frac{p(x)}{p(x)e^{\sin(x)}} = \lim_{x \rightarrow \infty} e^{-\sin(x)} \quad (33)$$

Define two sequences as  $n \rightarrow \infty$ .

$$a_n = \left(2n + \frac{1}{2}\right)\pi; b_n = \left(2n + \frac{3}{2}\right)\pi$$

Apply both  $a_n$  and  $b_n$  to  $e^{-\sin(x)}$ .

$$e^{-\sin(a_n)} = e^{-\sin((2n+\frac{1}{2})\pi)} = e^{-1} \quad (34)$$

$$e^{-\sin(b_n)} = e^{-\sin((2n+\frac{3}{2})\pi)} = e^1 = e \quad (35)$$

This gives us two subsequences that converge to two different values as  $x \rightarrow \infty$ . Therefore, the limit does not exist. OEΔ

We prove that  $\lim_{x \rightarrow \infty} \frac{p'(x)}{q'(x)} = 0$ .

*Proof.* First, we find the derivative of  $p(x)$  and  $q(x)$ . This involves the multiplication rule  $((fg)' = fg' + f'g)$ , the chain rule  $((f(g))' = g'f'(g))$ , my assertion that  $(\cos(x))' = -\sin(x)$ , and the  $\frac{d}{dx}e^x = e^x$ . I was given verbal approval in Office Hours to assert that  $\frac{d}{dx}e^x = e^x$ .

$$p'(x) = (x + \sin(x) \cos(x))' = 1 + (\sin(x) \cos(x))' = 1 + \cos^2(x) - \sin^2(x) \quad (36)$$

$$= 1 + \cos(2x) = 2 \cos^2(x) \quad (37)$$

$$q'(x) = \left(p(x)e^{\sin(x)}\right)' = p'(x)e^{\sin(x)} + p(x) \left(e^{\sin(x)}\right)' \quad (38)$$

$$= p'(x)e^{\sin(x)} + p(x) \cos(x)e^{\sin(x)} \quad (39)$$

$$= 2 \cos^2(x)e^{\sin(x)} + (x + \sin(x) \cos(x)) \cos(x)e^{\sin(x)} \quad (40)$$

$$= e^{\sin(x)} \cos(x) (2 \cos(x) + x + \sin(x) \cos(x)) \quad (41)$$

Now, we compute  $\frac{p'(x)}{q'(x)}$ .

$$\frac{p'(x)}{q'(x)} = \frac{2 \cos^2(x)}{e^{\sin(x)} \cos(x) (2 \cos(x) + x + \sin(x) \cos(x))} \quad (42)$$

$$= \frac{2 \cos(x)}{e^{\sin(x)} (2 \cos(x) + x + \sin(x) \cos(x))} \quad (43)$$



Take a few inequalities based on the bounds of  $\sin(x)$  and  $\cos(x)$  being  $-1 \leq \sin(x) \leq 1$  and  $-1 \leq \cos(x) \leq 1$ .

$$e^{-1} \leq e^{\sin(x)} \leq e \quad (44)$$

$$-2 \leq 2 \cos(x) \leq 2 \quad (45)$$

$$-1 \leq \sin(x) \cos(x) \leq 1 \quad (46)$$

$$-3 \leq 2 \cos(x) + \sin(x) \cos(x) \leq 3 \quad (47)$$

$$x - 3 \leq 2 \cos(x) + x + \sin(x) \cos(x) \leq x + 3 \quad (48)$$

$$e^{-1} (x - 3) \leq e^{\sin(x)} (2 \cos(x) + x + \sin(x) \cos(x)) \leq e (x + 3) \quad (49)$$

$$\frac{1}{e^{-1} (x - 3)} \geq \frac{1}{e^{\sin(x)} (2 \cos(x) + x + \sin(x) \cos(x))} \geq \frac{1}{e (x + 3)} \quad (50)$$

OEΔ

### 3 Problem 3

Let  $f : (a, b) \rightarrow \mathbb{R}$  be continuous. Let  $c \in (a, b)$ , and suppose  $f$  is differentiable on  $(a, c)$  and  $(c, b)$ . Suppose that

$$\lim_{x \rightarrow c} f'(x)$$

exists and is a real number. Show that  $f$  is differentiable at  $c$ . *Hint:* Mean value theorem.

#### 3.1 Solution

*Proof.* Let  $\lim_{x \rightarrow c} f'(x) = L$ . Recall the definition of  $f'(c)$ .

$$f'(c) = \lim_{t \rightarrow c} \frac{f(t) - f(c)}{t - c} \quad (51)$$

Without loss of generality, suppose that  $t > c$ . By the Mean Value Theorem, there exists  $d \in (c, t)$  where  $\frac{f(t) - f(c)}{t - c} = f'(d)$ . Since  $d \in (c, t)$ ,  $c < d < t$ , so  $0 < d - c < t - c$  and therefore  $0 < |d - c| < |t - c|$ .

$f'(c) = \lim_{t \rightarrow c} \frac{f(t) - f(c)}{t - c}$  means that for all  $\varepsilon > 0$ , there exists  $\delta > 0$  where  $|t - c| < \delta$  implies  $\left| \frac{f(t) - f(c)}{t - c} - f'(c) \right| < \varepsilon$ . Since  $|d - c| < |t - c|$ ,  $|d - c| < \delta$ . Furthermore, since  $\frac{f(t) - f(c)}{t - c} = f'(d)$ ,  $|f'(d) - f'(c)| < \varepsilon$ . Therefore, for all  $\varepsilon > 0$ , there exists  $\delta > 0$  where  $|d - c| < \delta$  implies  $|f'(d) - f'(c)| < \varepsilon$ . This means  $f'(c) = \lim_{d \rightarrow c} f'(d)$ . It is given that  $\lim_{x \rightarrow c} f'(x)$  exists and is a real number. Therefore  $f'(c)$  exists and is a real number. Ergo,  $f$  is differentiable at  $c$ .  $\square$

## 4 Problem 4

Let  $f : \mathbb{R} \rightarrow \mathbb{R}$  be  $C^2$  (meaning its second derivative exists and is continuous on  $\mathbb{R}$ ).

- (a) Let  $x \in \mathbb{R}$ . For each  $h > 0$ , show there are numbers  $\xi$  and  $\eta$  with  $x - h < \xi < x < \eta < x + h$  and

$$\frac{f''(\xi) + f''(\eta)}{2} = \frac{f(x+h) + f(x-h) - 2f(x)}{h^2}.$$

- (b) Show that, for each  $x \in \mathbb{R}$ ,

$$f''(x) = \lim_{h \rightarrow 0} \frac{f(x+h) + f(x-h) - 2f(x)}{h^2}.$$

- (c) A function  $f : \mathbb{R} \rightarrow \mathbb{R}$  is called *convex* if for all  $x, y \in \mathbb{R}$  and  $0 \leq \lambda \leq 1$ , we have

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y).$$

Geometrically, this means the graph of  $f$  lies *below* all of its secant lines. Show that any  $C^2$  convex function  $f$  satisfies  $f''(x) \geq 0$  for all  $x$ . *Hint:* convexity tells you something about the numerator in part (b).

- (d) Conversely, if a twice-differentiable function  $f$  satisfies  $f''(x) \geq 0$  for all  $x$ , show that  $f$  is convex. *Hint:* For  $\lambda \in (0, 1)$  and  $x < y \in \mathbb{R}$ , the number  $z := \lambda x + (1 - \lambda)y$  satisfies  $x < z < y$ . Apply the mean value theorem on  $[x, z]$  and  $[z, y]$  to get an expression for  $\lambda f(x) + (1 - \lambda)f(y)$ .

### 4.1 Solution (a)

*Proof.* Take the latter term and separate it so each new term has only one  $f(x)$ .

$$\frac{f(x+h) + f(x-h) - 2f(x)}{h^2} = \frac{f(x+h) - f(x)}{h^2} + \frac{f(x-h) - f(x)}{h^2} \quad (52)$$

By 0.4: *Alternate Mean Value Theorem*, there exists  $\alpha \in (x-h, x)$  where  $f'(\alpha) = \frac{f(x) - f(x-h)}{h}$  and there exists  $\beta \in (x, x+h)$  where  $f'(\beta) = \frac{f(x+h) - f(x)}{h}$ . We can put these in.

$$\frac{f(x+h) + f(x-h) - 2f(x)}{h^2} = \frac{f'(\beta)}{h} - \frac{f'(\alpha)}{h} = \frac{f'(\beta) - f'(\alpha)}{h} \quad (53)$$

Since  $\alpha \in (x-h, x)$ , either  $\alpha - \frac{h}{2} \in [x-h, x]$  or  $\alpha + \frac{h}{2} \in [x-h, x]$ , or both. Since  $\beta \in (x, x+h)$ , either  $\beta - \frac{h}{2} \in [x, x+h]$  or  $\beta + \frac{h}{2} \in [x, x+h]$ , or both. Without loss of generality, let  $\alpha + \frac{h}{2} \in [x-h, x]$  and  $\beta - \frac{h}{2} \in [x, x+h]$ . By the alternate mean value theorem, there exists  $\xi \in (\alpha, \alpha + \frac{h}{2})$  where  $f''(\xi) = \frac{f'(\alpha + \frac{h}{2}) - f'(\alpha)}{\frac{h}{2}}$  and there exists  $\eta \in (\beta - \frac{h}{2}, \beta)$  where  $f''(\eta) = \frac{f'(\beta) - f'(\beta - \frac{h}{2})}{\frac{h}{2}}$ . Add and subtracting  $\frac{f'(\beta - \frac{h}{2})}{h}$  and  $\frac{f'(\alpha + \frac{h}{2})}{h}$  from both sides of our earlier equation and apply our values of  $f''(\xi)$  and  $f''(\eta)$ .

$$\frac{f'(\beta) - f'(\alpha)}{h} = \frac{f'(\beta) + f'(\beta - \frac{h}{2}) - f'(\beta - \frac{h}{2}) + f'(\alpha + \frac{h}{2}) - f'(\alpha + \frac{h}{2}) - f'(\alpha)}{h} \quad (54)$$

$$= \frac{1}{2} \left( \frac{f'(\beta) - f'(\beta - \frac{h}{2})}{\frac{h}{2}} + \frac{f'(\alpha + \frac{h}{2}) - f'(\alpha)}{\frac{h}{2}} \right) + \frac{f'(\beta - \frac{h}{2}) - f'(\alpha + \frac{h}{2})}{h} \quad (55)$$

$$= \frac{f''(\xi) + f''(\eta)}{2} + \frac{f'(\beta - \frac{h}{2}) - f'(\alpha + \frac{h}{2})}{h} \quad (56)$$

OEΔ

## 5 Problem 5

Let  $f : [0, 1] \rightarrow \mathbb{R}$  be defined by

$$f(x) = \begin{cases} 1, & \text{if } x \in \mathbb{Q} \\ 0, & \text{if } x \notin \mathbb{Q}. \end{cases}$$

Compute the upper and lower Riemann integrals of  $f$ . Is  $f$  Riemann integrable?

### 5.1 Solution

The issue here is the density of the rationals in the reals.

First we prove the upper Riemann integral.

$$\overline{\int_0^1} f(x) dx = 1 \quad (57)$$

*Proof.* Let  $P$  be any partition of  $[0, 1]$  with  $n + 1$  elements. Take any  $j \in \mathbb{N}$  where  $1 \leq j \leq n$ . By the density of the rationals in the reals, there exists  $x \in [x_{j-1}, x_j]$  where  $x \in \mathbb{Q}$ . Since the only two possible values of  $f(x)$  are 0 and 1,  $\sup_{x \in [x_{j-1}, x_j]} f(x) = 1$ . We can compute the upper sum of  $f$  from here.

$$U(P, f) = \sum_{j=1}^n \Delta x_j \sup_{t \in [x_{j-1}, x_j]} f(t) = \sum_{j=1}^n \Delta x_j * 1 = \sum_{j=1}^n \Delta x_j \quad (58)$$

Since the sum of  $\Delta x_j$  telescopes,  $\sum_{j=1}^n \Delta x_j = x_n - x_0$ . Therefore, for any partition  $P$  of  $[0, 1]$ ,  $U(P, f) = 1 - 0 = 1$ . Since  $U(P, f) = 1$  for every partition  $P$  of  $[0, 1]$ ,  $\inf\{U(P, f) | \text{Partition } P\} = 1$ . Ergo,  $\overline{\int_0^1} f(x) dx = \inf\{U(P, f) | \text{Partition } P\} = 1$ . OEΔ

Now we prove the lower Riemann integral.

$$\underline{\int_0^1} f(x) dx = 0 \quad (59)$$

*Proof.* Let  $P$  be any partition of  $[0, 1]$  with  $n + 1$  elements. Take any  $j \in \mathbb{N}$  where  $1 \leq j \leq n$ . By the density of the irrationals in the reals, there exists  $t \in [x_{j-1}, x_j]$  where  $t \notin \mathbb{Q}$ . Since the only two possible values of  $f(x)$  are 0 and 1,  $\inf_{t \in [x_{j-1}, x_j]} f(t) = 0$ . We can compute the upper sum of  $f$  from here.

$$L(P, f) = \sum_{j=1}^n \Delta x_j \inf_{t \in [x_{j-1}, x_j]} f(t) = \sum_{j=1}^n \Delta x_j * 0 = \sum_{j=1}^n 0 = 0 \quad (60)$$

Therefore, for any partition  $P$  of  $[0, 1]$ ,  $L(P, f) = 0$ . Since  $L(P, f) = 0$  for every partition  $P$  of  $[0, 1]$ ,  $\sup\{L(P, f) | \text{Partition } P\} = 0$ . Ergo,  $\underline{\int_0^1} f(x) dx = \sup\{L(P, f) | \text{Partition } P\} = 0$ . OEΔ

Since  $\overline{\int_0^1} f(x) dx \neq \underline{\int_0^1} f(x) dx$ , the function is not Riemann integrable on  $[0, 1]$ .